USB POWER / ALL-RAIL PROTECTION AND RESET

S7: S6 circuit with the second output header J3 removed. All three converters retained; main and + rails TPS16630; negative LM5067 + MOSFET.

USB PD and main eFuse

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A nominal

File: 01_usb_input.kicad_sch

+12 V converter and output eFuse

MAIN_OK
P12_OK
RUN
V3V3
VIN
+12V
P12_RAW

TPS62933F -> TPS16630
+12 V protected output

File: 02_positive_12v.kicad_sch

+5 V converter and output eFuse

MAIN_OK
P5_OK
RUN
V3V3
VIN
+5V

TPS62933F -> TPS16630
+5 V protected output

File: 03_positive_5v.kicad_sch

Negative converter - raw rail

P12_RAW
VIN
N12_RAW

LMR38020F inverting buck-boost
RAW negative supply

File: 04_negative_converter.kicad_sch

Negative rail protection and level shifting

N12_RAW
NOT_RUN
V3V3
VIN
−12V
N12_STATUS

LM5067 + external N-MOSFET
1.0 A nominal / optical control

File: 05_negative_protection.kicad_sch

Protected outputs - voltage monitors and panel LEDs

+12V
+5V
−12V
MAIN_LED_OK
N12_STATUS
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK

One Eurorack output header (S7)
MAIN, +12, +5 and −12 LEDs

File: 07_outputs_status.kicad_sch

Always-on control - all-rail fault latch and RESET

MAIN_OK
N12_OK
NOT_RUN
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / retained ARMED + FAULT
Manual RESET disables all three rails

File: 06_fault_reset.kicad_sch

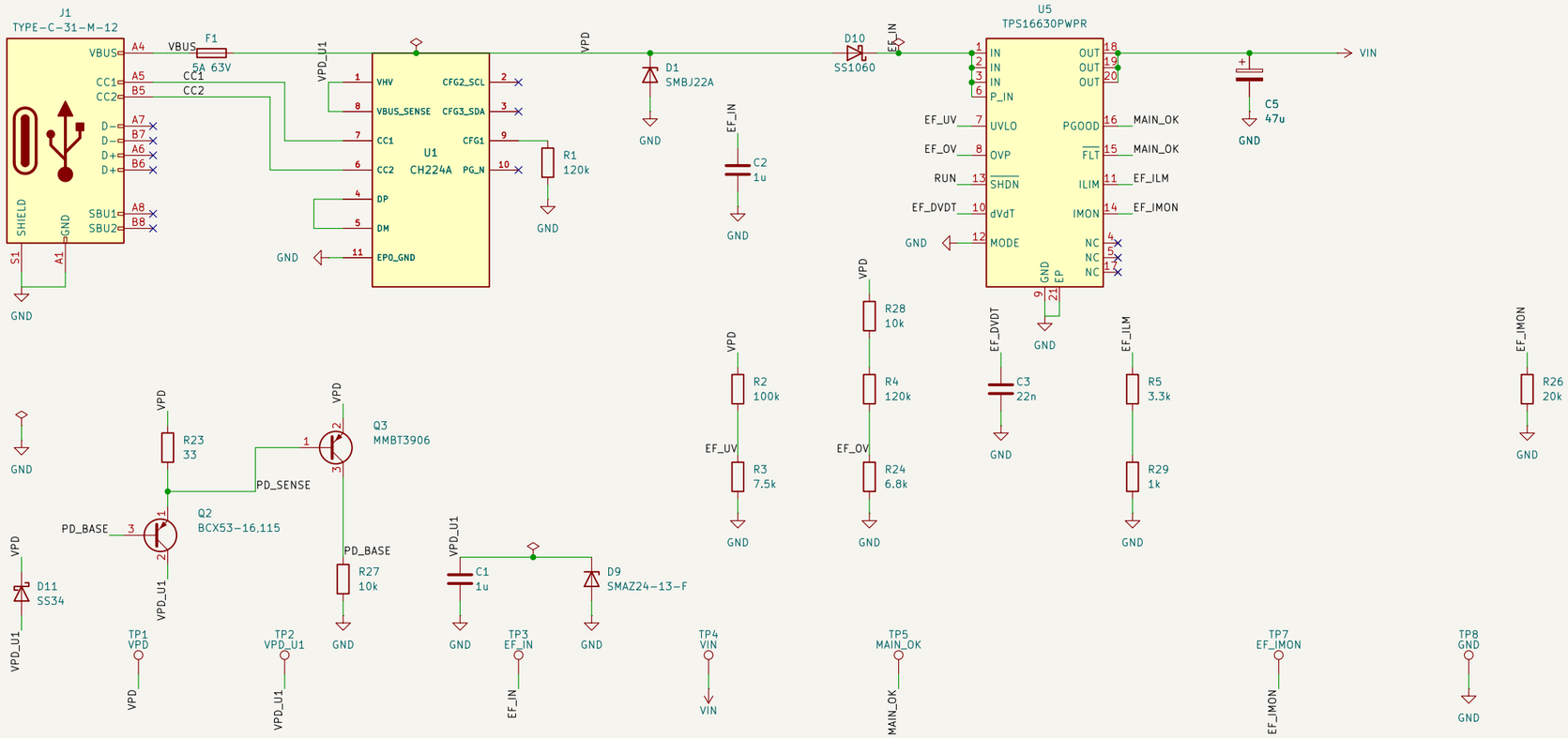
FAULT POLICY
Each rail detects independently.
Persistent fault -> all OFF until RESET.
Main LED is not merely a +5 V lamp.

OUTPUTS: +12 / +5 / −12
Nominal limits: 2.647 / 2.647 / 1.0 A
Tolerance and thermal limits apply.
Not a guaranteed 54.5 W simultaneous rating.

Design notes and budget: docs/design.md
S7: S6 per-rail protection, fault latch/reset, four LEDs; one output header (J3 removed)
Big Modular

Sheet: /	
File: usb-power-45w.kicad_sch	
Title: USB power - wired power flow and protection overview	
Size: User	Date:
KiCad E.D.A. 10.0.6	**Rev: S7**
	Id: 1/8

USB PD AND MAIN EFUSE



20 V PD request: U5 TPS16630 remains 60 V, 4.186 A nominal. RUN from retained fault latch.
MAIN_OK is wired FLT + PGOOD at 3.3 V, NOT a raw-voltage logic output.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇— MAIN_OK

RUN ◇— RUN

VIN ◇— VIN

VPD ◇— VPD

S7 — see design.md, s6-rail-protection.md and stack-r2.md

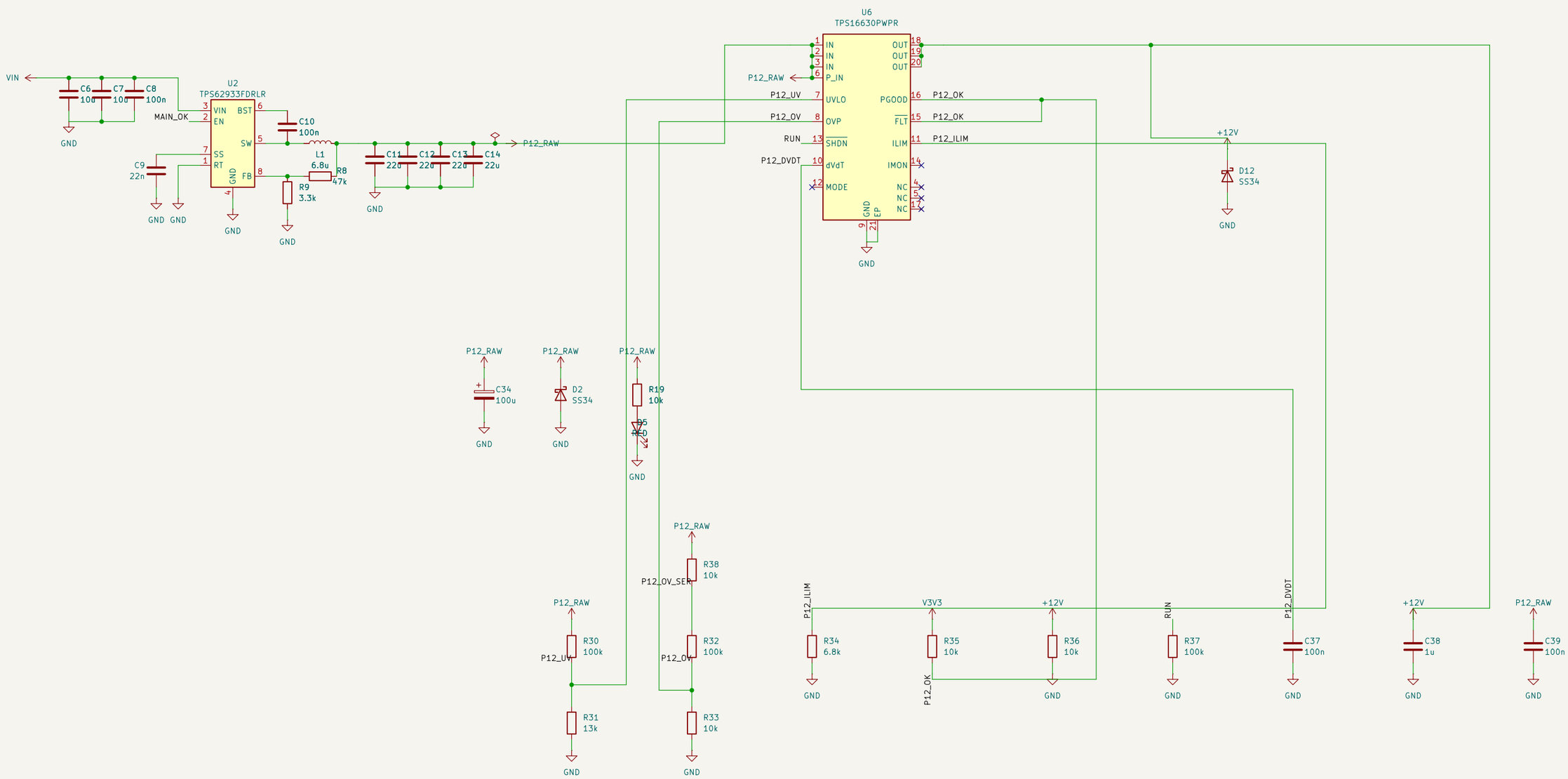
Big Modular

Sheet: /USB PD and main eFuse/
File: 01_usb_input.kicad_sch

Title: USB PD and main eFuse

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 2/8

+12 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. RAW feedback/bulk stay BEFORE U6.
U6: 2.647 A nominal, 2.335–2.971 A screened. No guaranteed 2.5 A minimum; MODE latch-off.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇—— +12V

MAIN_OK ◇—— MAIN_OK

P12_OK ◇—— P12_OK

P12_RAW ◇—— P12_RAW

RUN ◇—— RUN

V3V3 ◇—— V3V3

VIN ◇—— VIN

S7 — see design.md, s6-rail-protection.md and stack-r2.md

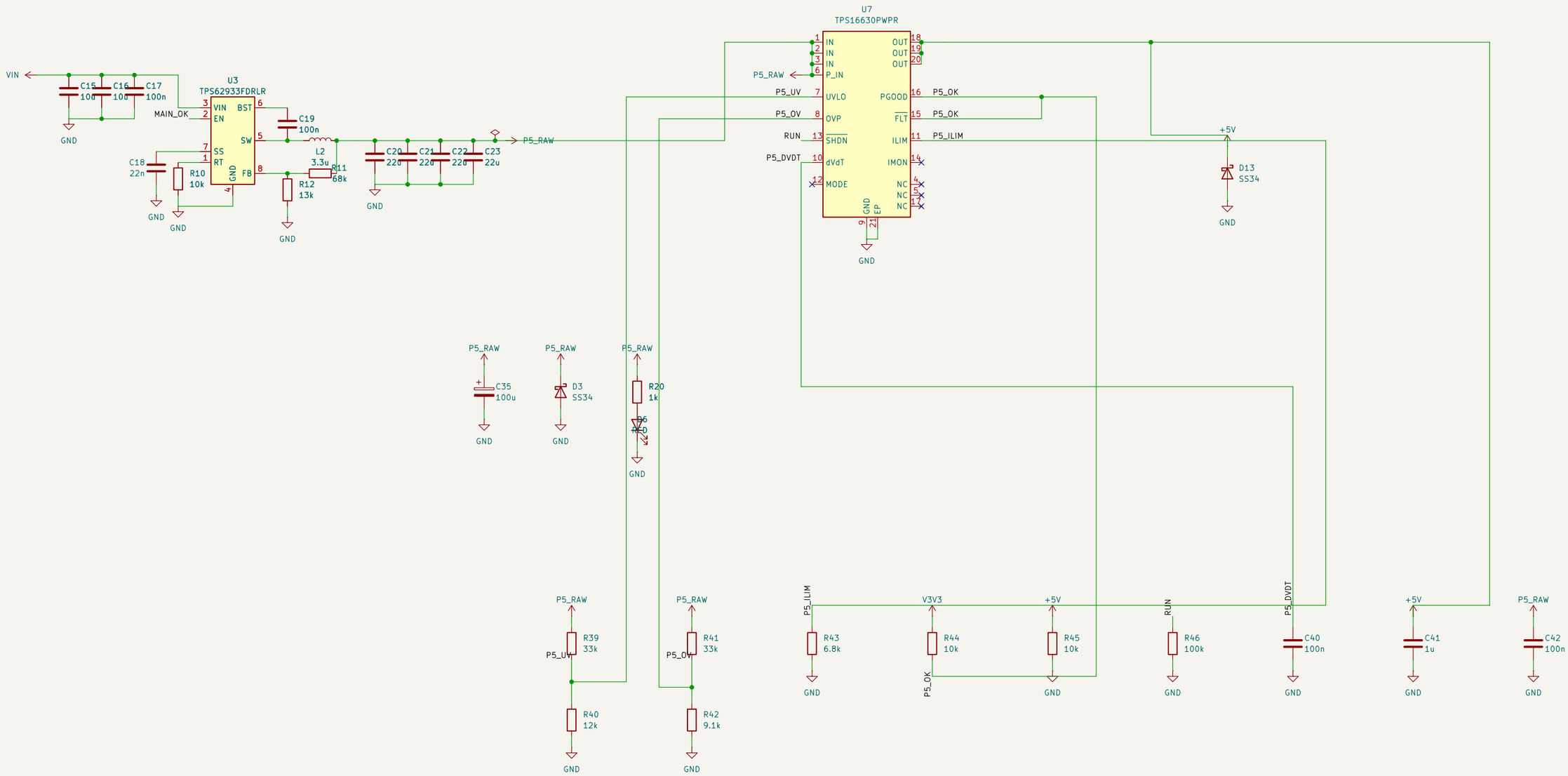
Big Modular

Sheet: /+12 V converter and output eFuse/
File: 02_positive_12v.kicad_sch

Title: +12 V converter and output eFuse

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 3/8

+5 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. U7 input must remain >=4.5 V.
U7: 2.647 A nominal, 2.335–2.971 A screened. Output-valid and FLT/PGOOD share P5_OK.

INTER-SHEET INTERFACE (see wired power-flow overview)

+5V ◇ +5V

V3V3 ◇ V3V3

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

P5_OK ◇ P5_OK

RUN ◇ RUN

S7 — see design.md, s6-rail-protection.md and stack-r2.md

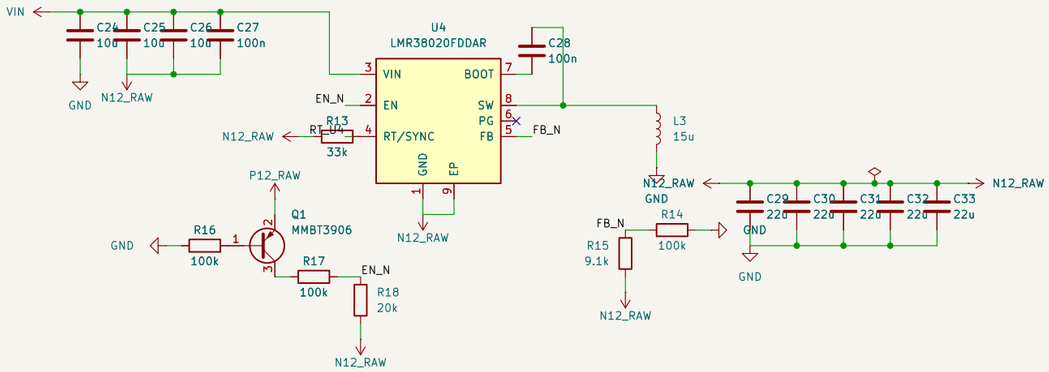
Big Modular

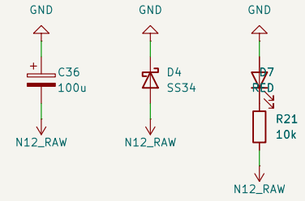
Sheet: /+5 V converter and output eFuse/
File: 03_positive_5v.kicad_sch

Title: +5 V converter and output eFuse

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 4/8

NEGATIVE CONVERTER — RAW RAIL





LMR38020F ground/exposed pad = N12_RAW, not system GND.
Q1 follows P12_RAW: load reaches N12_RAW ONLY through page 05 series protection.

INTER-SHEET INTERFACE (see wired power-flow overview)

N12_RAW ◇ N12_RAW

P12_RAW ◇ P12_RAW

VIN ◇ VIN

S7 — see design.md, s6-rail-protection.md and stack-r2.md

Big Modular

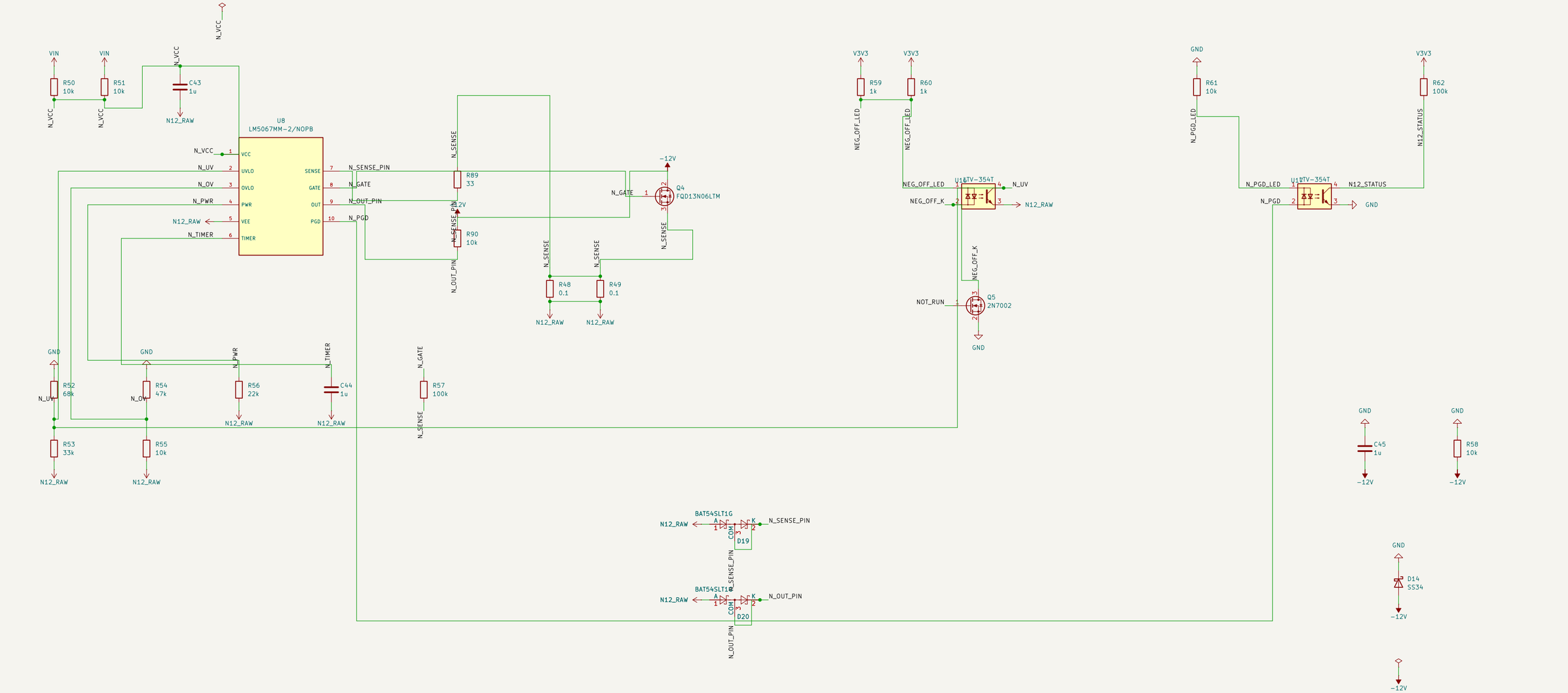
Sheet: /Negative converter — raw rail/
File: 04_negative_converter.kicad_sch

Title: Negative converter — raw rail

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 5/8

NEGATIVE RAIL PROTECTION AND LEVEL SHIFTING

System GND is continuous. R48 || R49 = 50 mOhm; Q4 is in the NEGATIVE conductor.
U8 VCC is resistor-fed from VIN. U16 disables via UVLO. U17 level shifts PGD.
Nominal 1 A / 3.1 W MOSFET limit. Reverse-collapse pin clamps do not block reverse power.



INTER-SHEET INTERFACE (see wired power-flow overview)

−12V −12V

N12_RAW N12_RAW

N12_STATUS N12_STATUS

NOT_RUN NOT_RUN

V3V3 V3V3

VIN VIN

S7 — see design.md, s6-rail-protection.md and stack-r2.md

Big Modular

Sheet: /Negative rail protection and level shifting/
File: 05_negative_protection.kicad_sch

Title: Negative rail protection and level shifting

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 6/8

ALWAYS-ON CONTROL — ALL-RAIL FAULT LATCH AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK

U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
R64 10k
R67 10k
R68 10k
C48 100n

U11
TPS3808G01DBVR
R69 33k
R70 10k
C49 1u
R71 10k
C50 100n

SW1
TL3340AF160QG

U12
SN74HC74DR

U13
SN74HC08DR

U14
SN74HC20DR
R72 10k
C51 100n

U15
SN74LVC1G17DBVR

C52 100n
C53 100n
C54 100n
C55 100n

TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP13 P5_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3

RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
Startup qualification ~5.7 s typical, then any persistent bad status latches OFF until RESET.
ARMED sheet-to-sheet signal (see wires & power-input connections).
MAIN_OK is a local rail-good flag (see wires & power-input connections). Local protection is active at startup.

MAIN_LED_OK
P5_OK
MAIN_OK
RUN
N12_OK
V20_OK
NOT_RUN
V3V3
P12_OK
VPD
TP6 FAULT_LATCH

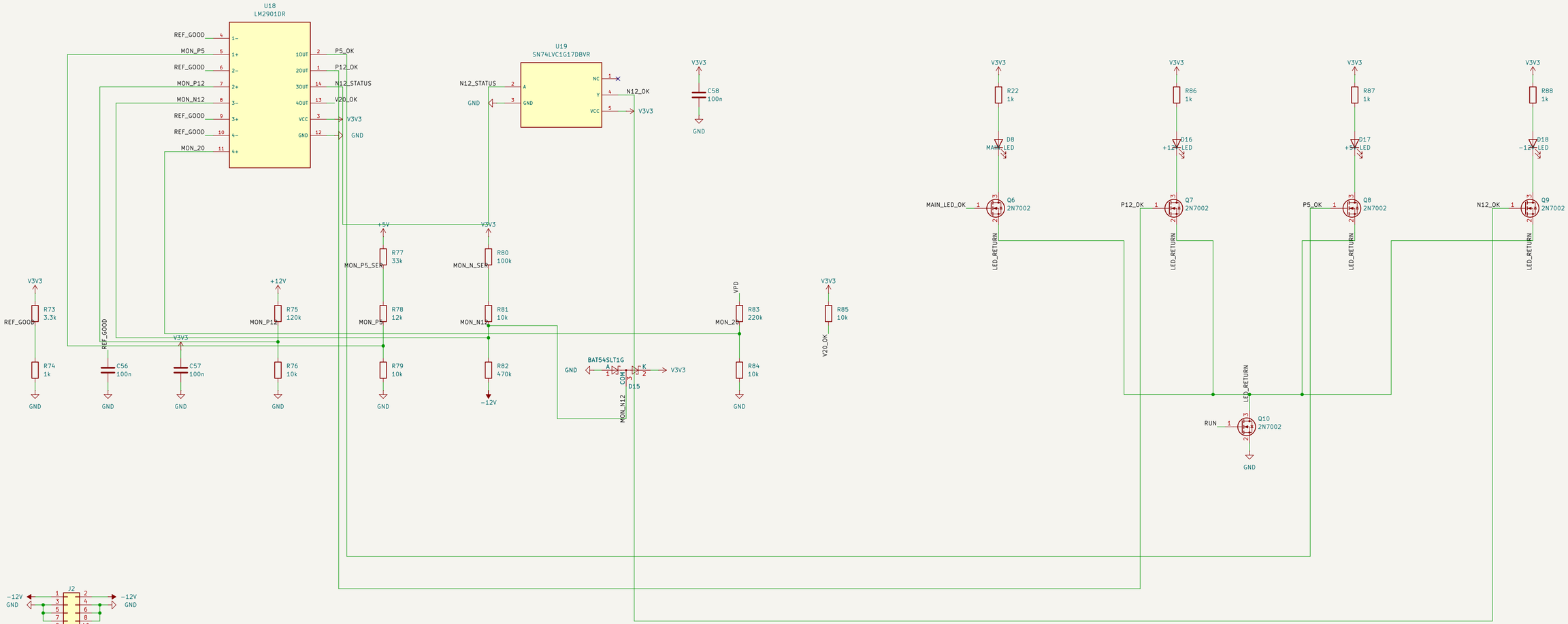
S7 — see design.md, s6-rail-protection.md and stack-r2.md
Big Modular
Sheet: /Always-on control — all-rail fault latch and RESET/
File: 06_fault_reset.kicad_sch

Title: Always-on control — all-rail fault latch and RESET

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 7/8

PROTECTED OUTPUTS — VOLTAGE MONITORS AND PANEL LEDS

D8 MAIN: input-valid AND main-eFuse status AND RUN. D16/+12, D17/+5, D18/−12: delivered-rail status AND RUN.
Any latched common fault turns all four panel LEDs OFF. D5/D6/D7 on converter pages indicate RAW rails.
Voltage-present thresholds are not +/−5% regulation guarantees: retained external charge is not actively discharged.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V
+5V — +5V
−12V — −12V
MAIN_LED_OK — MAIN_LED_OK
N12_OK — N12_OK
N12_STATUS — N12_STATUS
P12_OK — P12_OK
P5_OK — P5_OK
RUN — RUN
V20_OK — V20_OK
V3V3 — V3V3
VPD — VPD

S7 — see design.md, s6-rail-protection.md and stack-r2.md
Big Modular
Sheet: /Protected outputs — voltage monitors and panel LEDs/
File: 07_outputs_status.kicad_sch

Title: Protected outputs — voltage monitors and panel LEDs

Size: User	Date:	**Rev: S7**
KiCad E.D.A. 10.0.6		Id: 8/8